Mingxing Rao

Curriculum Vitae

mingxing.rao@vanderbilt.edu
mx-ethan-rao.github.io
mingxing-ethan-rao-488030261
mx-ethan-rao
nAJns40AAAAJ&hl

Education

2023–Present **Ph.D. in Computer Science**, *Vanderbilt University*, Nashville, USA
Advisor: Daniel Moyer; GPA: 3.8/4.0

2018–2022 **B.Sc. (Hons.) in Computer Science**, *Hong Kong Baptist University*, Hong Kong, China
Top 1/100; GPA: 3.7/4.0

Research Interests

Generative Models, Model Inversion, Membership Inference, Monocular Depth Estimation Representation Learning

Selected Publications

M. Rao, D. Moyer. *Generalization and Memorization in Rectified Flow*. Under review at **ECCV 2026**. Preprint: `https://arxiv.org/pdf/2603.13421v1`

M. Rao, B. Qu, D. Moyer. *Latent Diffusion Inversion Requires Understanding the Latent Space*. Published at **CVPR 2026**. Preprint: `https://arxiv.org/pdf/2511.20592`

M. Rao, B. Qu, D. Moyer. Submitted to **Neurocomputing**. *Score-based Membership Inference on Diffusion Models*. Preprint: `https://arxiv.org/pdf/2509.25003`

M. Rao, B. Jiang, D. Moyer. *Training Noise Token Pruning*. Preprint: `https://arxiv.org/pdf/2411.18092`

M. Rao, Y. Qin, S. Kolouri, J.Y. Wu, D. Moyer. *Zero-shot Prompt-based Video Encoder for Surgical Gesture Recognition*. **IJCARS 2024**; presented at **IPCAI 2024**

Z. Zhang†, **M. Rao**†, Y. Huang, L. Zhang. *Graph-Based Gaussian Mixture VAE Improves Metagenome Binning*. Poster at **RECOMB 2023**

Experience

2026 **Research Assistant**, *The Advanced Research Projects Agency for Health (ARPA-H)*, Nashville, US
- Work on the robotic vision development for an automated surgical program, specializing in Monocular Depth Estimation (MDE) for endoscopic imaging.
- Managed the end-to-end data pipeline, from collection to model architecture design, focusing on high-accuracy absolute depth perception during Central Airway Obstruction (CAO) and Benign Prostatic Hyperplasia (BPH) procedures.

2022 **Machine Learning Engineer Intern**, *Baidu NLP Group*, Shenzhen, China
- Developed protein–ligand binding simulation pipelines with Gromacs for data augmentation.
- Designed GNN model improving affinity prediction accuracy on PDBBind.

2021 **Research Intern**, *National University of Singapore*, Singapore
- Built academic knowledge graph: entity extraction, graph-based fusion, rule-based inference.
- Conducted blockchain data mining: built graphs from chain data, applied topic modeling to forums.

2021–2022 **Undergraduate Researcher**, *Hong Kong Baptist University*, Hong Kong, China
- Improved metagenomic contig binning using Gaussian Mixture VAEs with graph embeddings.
- Results published as co-first author in poster to **RECOMB 2023**.

Teaching

Teaching Assistant, Vanderbilt University (2023–2025): CS4262 Foundations of Machine Learning, CS4260 Artificial Intelligence, CS8395 Representation Learning, DS5460 Big Data Scaling

Skills

Programming: Python, PyTorch, TensorFlow, JAX, Java, C/C++, CUDA, Git, Linux, Docker
Languages: English, Chinese

Awards

2022 Scholastic Award (Top 2% academic performance)
2021 C.V. Starr Scholarship Fund (~35,000 HKD)
2021 Faculty of Science Overseas Research Scholarship for Outstanding Students in 2020-2021 (~30,000 HKD)
2019 Mr. & Mrs. Lau Chor Tak Scholarship Scheme for Outstanding Students (~75,000 HKD)